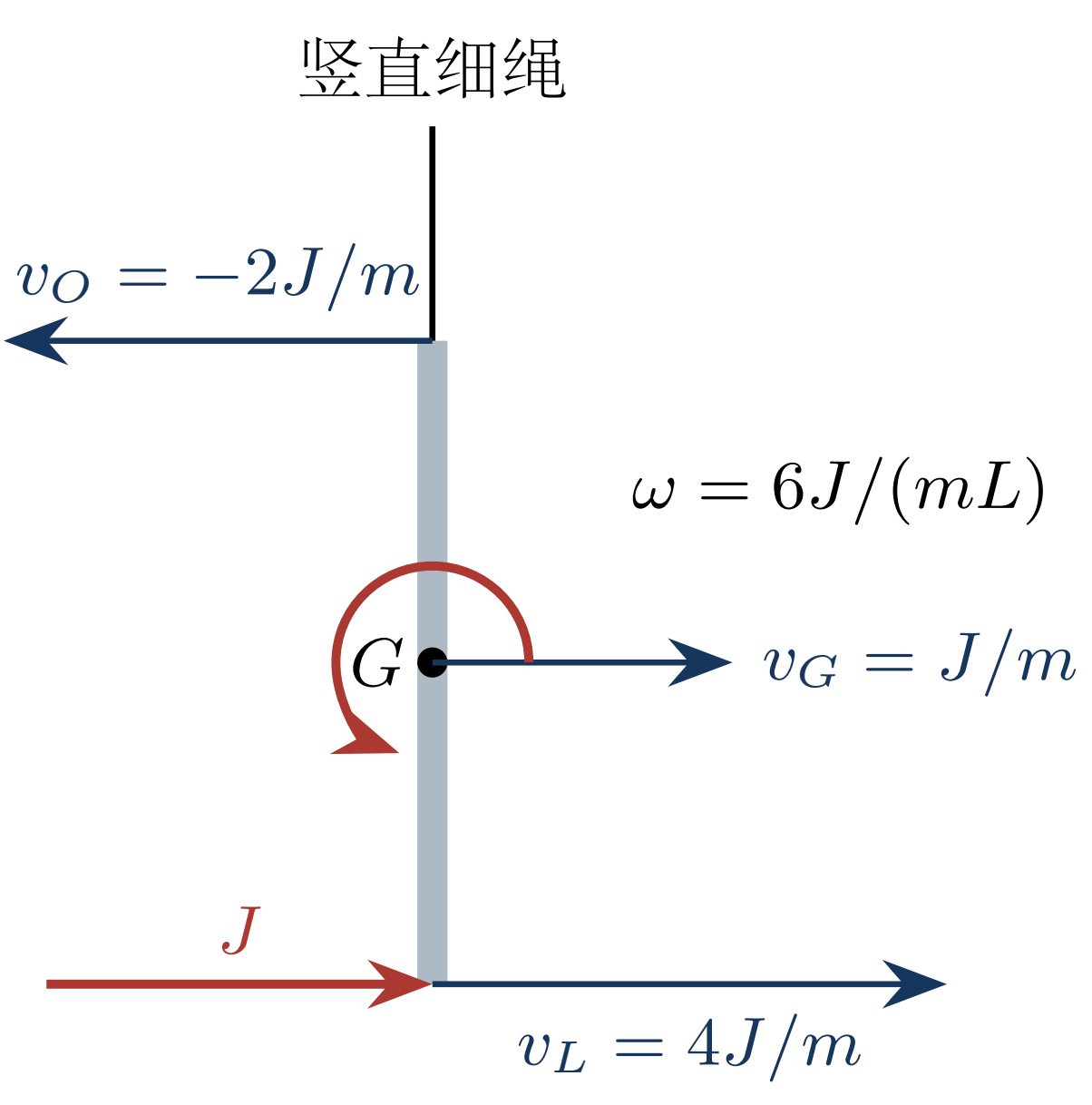
竖直细绳
$v_O = -2J/m$
$\omega = 6J/(mL)$
G
$v_G = J/m$
J
$v_L = 4J/m$